# Hurricane Harvey Hospital Flood Impacts: Accuracy of Federal Emergency Management Agency Flood Hazard Areas in Harris County, Texas

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*Objectives.* To compare the flood impacts experienced by Harris County, Texas, hospitals with Federal Emergency Management Agency (FEMA) flood hazard areas and Hurricane Harvey's inundation boundary.

*Methods.* One year following Hurricane Harvey, we created a novel data set of Hurricane Harvey's flood impacts in Harris County hospitals. We then mapped the hospital flood impact data in ArcGIS alongside FEMA flood hazard areas and Hurricane Harvey's inundation boundary to classify each hospital's location in high flood-risk areas and in areas purportedly affected by Hurricane Harvey.

*Results.* Of the 66 hospitals for which flood impact information was ascertained, 16 (24%) hospitals experienced flood impacts during Hurricane Harvey. Of these 16 hospitals, 5 (31%) were located outside a FEMA flood hazard area and 8 (50%) were located outside Hurricane Harvey's inundation boundary.

*Conclusions.* FEMA flood hazard areas did not accurately predict all areas of Harris County, Texas, that flooded during Hurricane Harvey or which hospitals experienced flood impacts. (*Am J Public Health.* 2020;110:574–579. doi:10.2105/AJPH.2019.305520)

Hurricane Harvey made landfall in Texas as a Category 4 hurricane on August 26, 2017. Areas of Harris County, Texas, received between 40 and 60 inches of rain, surpassing the previous single-storm rainfall record in the continental United States.[1] The ensuing flooding, influenced by a variety of factors, left an estimated 25% of Harris County underwater, with 50% of total flooding occurring outside Federal Emergency Management Agency (FEMA) flood hazard areas.[2–4]

Initial reports indicated that hospitals across Harris County scrambled to evacuate patients amid basement flooding and damage to kitchens, pharmacies, and supplies.[5] Although many of these hospitals had previous experience with flooding and recently fortified buildings to increase their preparedness, many also experienced difficulty withstanding Harvey's flood impacts: hospital access roads were blocked, winds prevented helicopter landings, and food supplies ran low.[5,6]

Hospitals are critical infrastructure that should be capable of delivering both day-to-day and emergency services at surge capacity during extreme weather events, such as flooding. Aside from providing care to currently admitted patients, hospitals must also manage new patients with flood-related health concerns; this necessitates efforts to minimize hospitals' flood-related complications related to power generation, clean water provision, patient safety, communication, and access.[7–9]

Planning for extreme flooding events and reducing hospital vulnerabilities require understanding where flood impacts are most likely. The FEMA National Flood Hazard Layer is a geospatial database used to map high flood-risk areas (100-year and 500-year flood hazard areas), but several recent studies have expressed concerns about its reliability for planning and policy purposes.[10,11] For hospitals, decisions on matters such as evacuation procedures and flood insurance can be based primarily on FEMA flood hazard areas. Therefore, it is important to determine this data source's validity in predicting where water inundation and impacts occurred from an extreme flooding event.

Harris County is located in southeast Texas and is highly susceptible to heavy rainfall and storm surge events because of its proximity to the Gulf of Mexico.[12,13] Harris County's dams and bayous, which are intended to hold and channel water toward the Gulf, are at risk of failing and are often overwhelmed by flood waters.[13] This physical vulnerability, paired with the explosive growth in Harris County from approximately 2 million to 6.5 million people since 1970, is putting an unprecedented number of people at risk during extreme flooding events.[13] Unchecked urban sprawl since the 1990s has resulted in the loss of almost 30% of the county's wetlands, which were largely responsible for retaining stormwater.[14,15] Although the Texas coast has experienced several extreme flooding events in the past 2 decades, including Tropical Storm Allison (2001), Hurricane Ike (2008), the Memorial Day Flood (2015), and the Tax Day Flood (2016), many questioned whether Harris County had adequately prepared for Hurricane Harvey's arrival in 2017.[15–17]

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Hurricane Harvey provided an opportunity to use geospatial analysis to validate FEMA flood hazard area data against the flood impacts experienced in Harris County's hospitals and the hurricane's inundation boundary, demonstrating the reliability of these data for future extreme flood planning in coastal Texas and elsewhere.

## METHODS

We focused on the 80 nonpsychiatric hospitals in Harris County, Texas. We excluded psychiatric hospitals from this analysis because they do not serve the general public or address physical health concerns during a hurricane.

### Data Sources

*Hospital flood impacts in Harris County, Texas.* We first created a novel data set of Hurricane Harvey's flood impacts experienced in Harris County hospitals through secondary data searches and telephone calls with hospitals. The "flood impacts" definition we used encompassed any case of water entering a hospital building. We used 2 methods to obtain information for as many hospitals as possible while also cross-validating our findings.

First, we searched for secondary data from news reports and documents from government agencies and local organizations. We identified these secondary sources through Google searches using hospital names and keywords such as "Hurricane Harvey" and "flood." We categorized the secondary sources used in this analysis as follows: news reports (n = 31), peer-reviewed journal articles (n = 1), government agency documents (n = 2), local organization documents (n = 3), and hospital Web sites (n = 10).

Next, we called each of the 80 hospitals during business hours and asked to speak with staff from facilities or emergency management to determine whether the hospital experienced any flooding during Hurricane Harvey. Using the combined results of the secondary source search and telephone calls with hospitals, we were able to ascertain flood impacts for 66 of the 80 hospitals in Harris County, leaving 14 hospitals with no data on flood impacts. We did not find any disagreements between the 2 methods used for obtaining hospital flood impacts.

*Geospatial data sets.* The shapefiles we used for the geographic information system analysis of flood impacts from Hurricane Harvey on hospital infrastructure are summarized here.

The Homeland Infrastructure Foundation-Level Data (HIFLD) hospitals shapefile contains all US hospitals' location information and descriptive characteristics acquired from various state or federal sources (https://hifld-geoplatform.opendata.arcgis.com/datasets/hospitals). We extracted HIFLD information for all hospital types except for psychiatric hospitals in Harris County, Texas, from April 2017. To map the novel hospital flood impact data set described earlier, we spatially linked them to the HIFLD hospitals shapefile.

We acquired Harris County's FEMA flood hazard areas from the National Flood Hazard Layer database (https://msc.fema.gov/portal/advanceSearch). This geospatial database contains polygons delineating high flood-risk areas (100- and 500-year flood hazard areas) that are based on current hydrology, infrastructure, and land use. The 100-year flood hazard areas delineate land with a 1% chance of flooding in any given year, whereas the 500-year flood hazard areas delineate land with a 0.2% chance of flooding in any given year. The FEMA National Flood Hazard Layer database is updated through several processes, including Letters of Map Revision, Letters of Map Amendment, and a cost–benefit approach, in which communities are chosen for updates based on criteria such as development levels and date of last update. Because the FEMA National Flood Hazard Layer database's historic versions are not archived online, we performed our analysis retrospectively using the January 2018 version.

We acquired 2 versions of Hurricane Harvey's inundation boundary, 1 from Harris County Flood Control District (HCFCD; https://www.hcfcd.org/hurricane-harvey/countywide-impacts) and the other from FEMA's Natural Hazard Risk Assessment Program (https://data.femadata.com/FIMA/NHRAP/Harvey). Both versions estimate the maximum area of water inundation during the hurricane. The HCFCD created an inundation boundary by assessing high-water marks using water-monitoring sites and manual surveying to identify where the bayous and tributaries overflowed their banks during Hurricane Harvey.[18]

The FEMA Natural Hazard Risk Assessment Program created another version of the inundation boundary by supplementing the initial HCFCD high-water marks with US Geological Survey high-water marks. For each data set, HCFCD and FEMA interpolated the high-water marks into a continuous water surface elevation using the inverse distance weighting method, and then they compared these with Harris County's digital elevation model to delineate the inundation's estimated boundary. Both versions of Hurricane Harvey's inundation boundary were created in 2017 following Hurricane Harvey.

To capture the full potential inundation extent of Hurricane Harvey, we merged the HCFCD and FEMA inundation extent shapefiles using a geometric union, which allowed all areas assessed as inundated in either data set to be considered inundated in the merged data. It is important to note that neither of these inundation boundary estimates necessarily shows all flood impacts throughout Harris County or necessarily indicates where structural flooding occurred, because water levels, as well as structure elevations and flood-resistant design, varied widely throughout the inundated areas.

### Geographic Information System Analysis

Our first goal was to identify the areas of overlap and nonoverlap between Harris County's FEMA flood hazard areas and Hurricane Harvey's inundation boundary, allowing an assessment of whether the FEMA flood hazard areas accurately predicted the areas of Harris County that would flood during an extreme flooding event such as Hurricane Harvey.

Our second goal was to assess whether Harris County hospital flood impacts from Hurricane Harvey were accurately predicted (1) by the hospitals' flood risk because of their locations in FEMA flood hazard areas or (2) by their locations in Hurricane Harvey's inundation boundary. To do this, first we classified each hospital's location as either inside or outside a FEMA flood hazard area, because the likelihood of hospital flood impacts should be higher for hospitals located in a FEMA flood hazard area. Next, we classified

each hospital's location as either inside or outside Hurricane Harvey's inundation boundary. We then compared these 2 flood-risk classifications with whether hospitals reported (either verbally in a telephone call or through secondary data) flood impacts during Hurricane Harvey based on the flood impacts data set we created as outlined.

We completed all geographic information system analyses in ArcMap version 10.6.1 (ESRI, Redlands, CA).[19]

## RESULTS

Using both primary and secondary data sources, we ascertained information about Hurricane Harvey's flood impacts for 66 of the 80 nonpsychiatric hospitals in Harris County included in this study. Of the 66 hospitals for which we ascertained flood impact information, 16 hospitals (24% of those ascertained) experienced flood impacts during Hurricane Harvey and 50 hospitals (76% of those ascertained) did not experience flood impacts during Hurricane Harvey. Table A (available as a supplement to the online version of this article at http://www.ajph.org) presents the flood impacts by hospital, as well as their locations inside or outside the FEMA flood hazard areas and Hurricane Harvey's inundation boundary.

The geographic information system analyses revealed considerable disagreement between the FEMA flood hazard areas and Hurricane Harvey's inundation boundary. Totaling 382 square miles, the areas of overlap between the FEMA flood hazard areas and Hurricane Harvey's inundation boundary, such as in the Addicks and Barker reservoirs and around Lake Houston and other major waterways, are shown in beige in Figure 1. However, we observed 2 types of nonoverlap. Approximately 37% (227 $mi^2$) of the FEMA flood hazard areas did not experience inundation from Hurricane Harvey according to Hurricane Harvey's inundation boundary (red represents both 100- and 500-year FEMA flood hazard areas in Figure 1). Approximately 18% (83 $mi^2$) of the areas that experienced inundation were located outside FEMA flood hazard areas (blue in Figure 1).

The 80 nonpsychiatric hospitals in Harris County included in this study are also mapped in Figure 1. The 16 hospitals that experienced flood impacts during Hurricane Harvey are shown as green dots, and the 50 hospitals that did not experience flood impacts during Hurricane Harvey are shown as black dots. Hospitals for which Hurricane Harvey flood impacts were not ascertained (n = 14) are represented as black Xs.

Next, we sought to understand whether the hospital flood impacts were predicted well by the FEMA flood hazard areas and Hurricane Harvey's inundation boundary. From top to bottom, the flowchart in Figure 2 displays whether hospitals in Harris County were located inside or outside a FEMA flood hazard area, followed by whether they were located inside or outside Hurricane Harvey's inundation boundary, followed by whether they experienced flood impacts during Hurricane Harvey. The results presented in the bottom row are categorized into 3 themes: expected results, unpredicted results with potential explanations, and unexpected flooding.

Of the 66 hospitals whose flood impacts we ascertained, 40 (60%) fell into the "expected results" theme (shown with a solid outline in Figure 2). We largely expected these hospitals' flood impacts (or lack thereof) based on the hospitals' flood risk. For example, 6 hospitals in this category experienced flood impacts, but this should have been expected because they had a high flood risk owing to their locations in a FEMA flood hazard area and Hurricane Harvey's inundation boundary. Similarly, 34 hospitals did not experience flood impacts, but this was expected because their locations were outside FEMA flood hazard areas and Hurricane Harvey's inundation boundary.

Of the 66 hospitals whose flood impacts we ascertained, 21 (32%) fell into the "unpredicted results with potential explanations" theme (shown with a dotted outline in Figure 2). These hospitals' flood impacts were largely unpredicted but have potential explanations worth exploring. For example, 16 hospitals did not experience flood impacts despite high flood risk owing to their locations inside both the FEMA flood hazard area and Hurricane Harvey's inundation boundary (n = 3) or locations inside a FEMA flood hazard area only (n = 13). Five other hospitals in this category did experience flood impacts, but we did not predict this because they were located outside Hurricane Harvey's inundation boundary (despite being inside the FEMA flood hazard area). We explore the potential explanations for these unpredicted results in the Discussion.

Of the 66 hospitals whose flood impacts we ascertained, 5 (8%) fell into the "unexpected flooding" theme (shown with a dashed outline in Figure 2). These hospitals' flood impacts were unexpected based on the low flood risk as a result of their locations outside a FEMA flood hazard area (2 hospitals were located inside Hurricane Harvey's inundation boundary, and the remaining 3 were located outside the inundation boundary). Considering their lower flood risk, it is possible that these hospitals were perhaps less prepared for the flood impacts.

## DISCUSSION

To our knowledge, our study is the first to document Hurricane Harvey's flood impacts on Harris County hospitals and compare these impacts to FEMA flood hazard areas through geospatial analysis. FEMA flood hazard areas are heavily considered in hospital emergency preparedness planning; therefore, it is imperative to establish whether the FEMA flood hazard areas are reliable in predicting hospital flood impacts.

Our analysis first revealed that the FEMA flood hazard areas did not accurately predict the areas of Harris County, Texas, that flooded during Hurricane Harvey in August 2017. The analysis also indicated that of the 16 hospitals that experienced Hurricane Harvey flood impacts, only 11 (69%) were located in a FEMA flood hazard area. Thus, the remaining 5 hospitals that experienced flood impacts were located outside a FEMA flood hazard area—a particularly concerning finding considering that these hospitals were likely less prepared because of their lower perceived risk.

Despite being used by hospitals to identify flood risk, plan emergency procedures, and inform policy decisions, the FEMA flood hazard area data had not previously been validated against the impacts of flooding on Harris County's hospital infrastructure. Our study demonstrates that FEMA flood hazard areas did not fully capture hospital flood impacts, with severe implications for health care service continuity and patient safety. Our concerns about the data's validity support

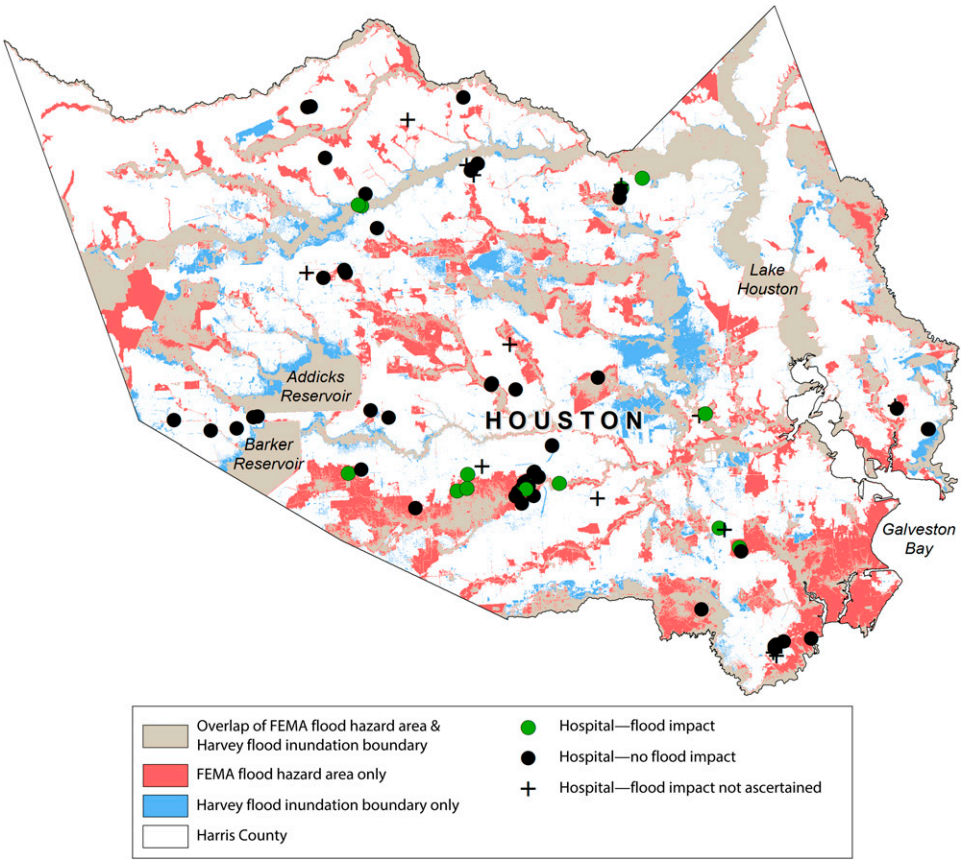


**FIGURE 1—Overlap and Nonoverlap Between Federal Emergency Management Agency (FEMA) Flood Hazard Areas and Hurricane Harvey's Inundation Boundary, and Hospital Locations and Flood Impacts Related to Hurricane Harvey: Texas, 2017**

several other studies' findings documenting disagreement between FEMA's flood-risk estimates and actual flood losses, particularly in coastal Texas.[10,11]

Although our results primarily indicate reliability concerns with the FEMA flood hazard areas, 2 other potential explanations exist for the unpredicted flood impacts found. First, some hospitals (n = 8) located outside Hurricane Harvey's inundation boundary still experienced flood impacts, potentially implying problems with the inundation boundary's accuracy. Alternatively, because our "flood impacts" definition was broad (encompassing any case of water entering a hospital building), it is possible that the reported hospital flood impacts were not attributable to ground-level water inundation, but instead structural problems that led to falling water getting into the hospital. Our study, however, was not able to confirm each hospital's flooding source. Second, some hospitals (n = 3) located inside Hurricane Harvey's inundation boundary did not experience flood impacts, suggesting that these hospitals may have prevented flooding by implementing flood-adaptation strategies in advance. We were unable to confirm either of these potential explanations, but it is crucial that both be further explored in future studies to inform hospital preparations for future extreme flooding in Harris County and elsewhere.

## Limitations and Future Directions

Some study limitations, as well as suggestions for forthcoming studies, are worth noting. First, because no historic versions of the FEMA flood hazard area data exist, we considered only the January 2018 version available during analysis. Considering that FEMA flood hazard area reevaluations take time, it is unlikely that the updated version already incorporated Hurricane Harvey's impact, and, therefore, we do not anticipate any drastic differences in the FEMA flood hazard area data between August 2017 and January 2018. Additionally, we considered only 2 versions of Hurricane Harvey's inundation boundary, those created by FEMA and HCFCD, neither of which considered floodwater depth. Future studies would benefit from comparing a larger number of inundation boundaries, including floodwater depth estimates, to understand the full extent of flooding related to a hurricane or other extreme rainfall event.

Second, our broad "flood impacts" definition may have unintentionally included impacts unrelated to ground-level water inundation, thus inflating the category of hospitals with unexpected flood impacts. However, these other flood impacts are

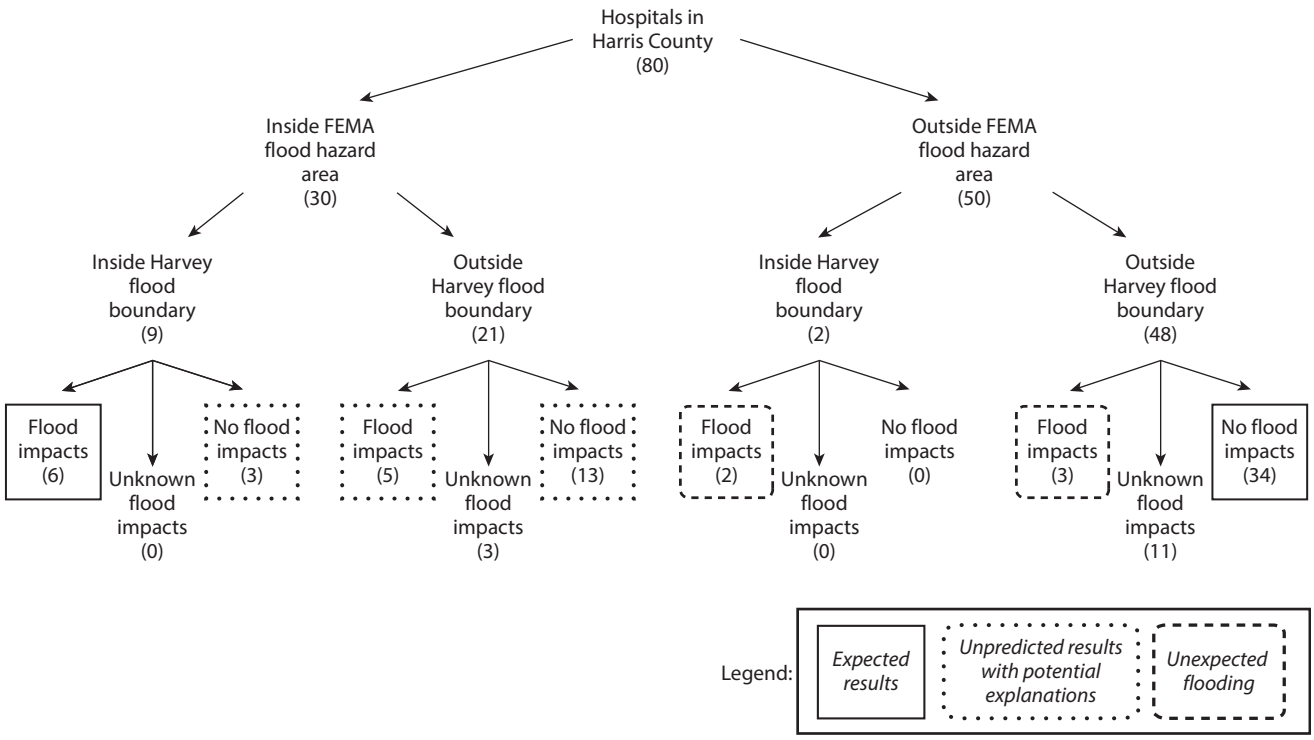


**FIGURE 2—Flowchart of Harris County Hospital Locations and Their Hurricane Harvey Flood Impacts: Texas, 2017**

important and should be further investigated. Additionally, hospitals with more severe flood impacts may be more likely to disclose their flood impacts, meaning our findings could underestimate Hurricane Harvey's true impacts on Harris County's hospitals. Future studies should use different flooding categories with specific definitions to separate out the causes of flooding and help inform how hospitals prepare for future flooding events.

Additionally, even though we used 2 methods to ascertain hospital flood impacts, we were unable to ascertain flood impacts for all 80 hospitals. We also limited our analysis to nonpsychiatric hospitals. Future studies on health care service flood impacts should include all facilities to determine if the impacts we uncovered apply more widely.

## Recommendations

Based on our findings, we recommend the following for flood-prone hospitals and their partner organizations in academia, private industry, and government. First, because our study's results are only applicable to Harris County, Texas, we advise that future studies validate FEMA flood hazard area data against flood extents and impacts in other US case studies. Second, because FEMA flood hazard areas are the traditional source of flood-risk estimates, we suggest exploring avenues to improve the FEMA flood hazard area data's reliability in predicting flood extents and impacts. Third, researchers should test whether alternative data sources could potentially complement, or even replace, FEMA flood hazard area data to inform emergency preparedness plans and policy decisions.

## Public Health Implications

Hospitals are a critical component of our country's health care infrastructure and are tasked with delivering both day-to-day and emergency services at surge capacity during extreme flooding events. Our results document a concerning inconsistency between FEMA flood hazard areas and hospitals' flood impacts following Hurricane Harvey in Harris County, Texas, strengthening the evidence base that FEMA flood hazard areas deserve further scrutiny in informing hospital emergency preparedness plans and policy. In the interest of health care service continuity and patient safety during extreme flooding events, hospitals nationwide must be provided with reliable flood-risk estimates.

Overall, this study contributes to efforts aimed at improving US hospitals' ability to better prepare for, respond to, and recover from future catastrophic flooding events such as Hurricane Harvey. With extreme flooding events predicted to become more frequent and intense in many regions across the United States over the next decade,[20] it is crucial to fully understand the risks and impacts of flooding on health care infrastructure to ensure health care availability and accessibility during future extreme flooding events.[7,9] AJPH

### CONTRIBUTORS

E. Hines conceptualized the study, collected and analyzed the data, and wrote the article. C. E. Reid supervised the study conceptualization and the data collection and analysis and critically revised the article. Both authors designed the research, discussed the results, and wrote the final version of the article.

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## CONFLICTS OF INTEREST

Neither author has any conflicts of interest to declare.

## HUMAN PARTICIPANT PROTECTION

The University of Colorado Boulder institutional review board deemed this study exempt from review because no human participants were involved in this study.

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